

CONFINED SPACE ENTRY

Control atmospheric hazards, energy sources and rescue risks.

PURPOSE AND SCOPE

Review oxygen deficiency or enrichment, toxic and flammable atmospheres, unexpected energy, engulfment, entrapment and delayed-rescue hazards, and verify that the permit, gas testing, isolation, ventilation, attendant and rescue arrangements are complete before entry. A confined space includes tanks, columns, reactors, ducts, pits, manholes, silos and similar areas with limited entry or exit, not designed for continuous occupancy and capable of containing atmospheric or physical hazards.

KEY HAZARDS

 OXYGEN	 TOXIC GAS	 FLAMMABLE ATMOSPHERE	 ENGULFMENT
 ENERGY	 RESTRICTED ACCESS	 LOST COMMUNICATION	 RESCUE

CRITICAL VALUES TO VERIFY IN THE FIELD

19.5-23.5% Acceptable oxygen range	<10% LEL General entry limit	100% Continuous monitoring during entry
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REMEMBER - SERNEM

- | | | |
|-------------------------|----------------------------|----------------------------------|
| S Stop and think | E Evaluate the risk | R Remove or reduce |
| N Never assume | E Ensure protection | M Monitor and communicate |

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STANDARDS AND FIELD REQUIREMENTS

OSHA 1910.146 requires space classification, a permit, atmospheric testing, an attendant, communication and rescue arrangements.

OSHA 1910.146(c)

SPACE EVALUATION

Identify confined spaces, mark permit spaces and prevent unauthorized entry.

OSHA 1910.146(d)

PERMIT PROGRAM

Manage evaluation, isolation, ventilation, testing, communication and rescue through a written program.

OSHA 1910.146(d)(5)

ATMOSPHERIC TESTING

Test oxygen first, then flammables and toxics with suitable equipment, and monitor during entry.

OSHA 1910.146(k)

RESCUE

Verify timely rescue capability and non-entry retrieval equipment where feasible.

MINIMUM EVIDENCE BEFORE START

- | | |
|---|---|
| ☐ Approved entry permit | ☐ Isolation / blinding list |
| ☐ Gas-test record | ☐ Monitor calibration / bump-test record |
| ☐ Attendant and communication plan | ☐ Rescue plan and equipment check |

SUPERVISOR DISCUSSION POINTS

- Is the confined-space entry permit approved and current?
- Who completed the latest gas test and what were the results?
- Which energy and process connections have been isolated?

PERMIT-REQUIRED CONFINED SPACE ENTRY SETUP



Isolation - Gas test - Ventilation - Attendant - Tripod - Rescue

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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. AVOID ENTRY

Verify the approved confined-space entry permit. Isolate the space and all connected energy sources.

2. ISOLATE AND VENTILATE

Test the atmosphere at upper, middle and lower levels. Provide continuous atmospheric monitoring where required. Provide adequate mechanical ventilation.

3. APPLY PERMIT AND MONITORING

Assign a competent and dedicated attendant. Maintain continuous communication with entrants. Record entry, exit and personnel count.

4. BE RESCUE READY

Prepare the rescue plan, team and equipment before entry. Evacuate immediately if conditions change.

CRITICAL WARNING



**Entering a confined space without a rescue plan can create a second casualty.
Raise the alarm and follow the approved rescue procedure.**

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|---|--|
| ☐ Entry permit approved | ☐ Risk assessment suitable |
| ☐ Isolation complete | ☐ Gas test acceptable |
| ☐ Detector checked | ☐ Ventilation ready |
| ☐ Attendant in position | ☐ Communication established |
| ☐ Rescue equipment ready | |



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ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			